

MARSH

Property Risk Evaluation Report

[Site Photo]

[ADDRESS]

[POSTAL]

[PROPERTY]

13 April, 2026

[CLIENT]	
Location Surveyed	[PROPERTY] [PROPERTY]
Surveyed By	[NAME], Senior Consultant, Marsh Risk Consulting
Survey Dates	26 February 2026
Date Last Surveyed	26 June 2022
Site Contacts	[NAME] – Building Manager, Facilities Management
Q/A Review By	-
File Name & Rev.	-

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Section 1

Purpose and Scope

Purpose

The purpose of this property risk evaluation report is to assist the property damage and business interruption insurance placement of [PROPERTY] LLP and/or [CLIENT]. It is made on request by Marsh Client Service Team to provide underwriting information and risk evaluation details for the [PROPERTY] [PROPERTY], hereinafter known as [PROPERTY] T3.

Scope

This risk evaluation was performed during the Marsh's 1-day site inspection to [PROPERTY] T3 on 26-February 2026, and involved a review of building construction, operations, fire protection systems & features and fire safety management & control. The review is not intended to define compliance with any specific standards and is limited to the topics identified.

This report is based upon conditions and practices observed and information made available to Marsh Advisory at the time of our survey. The report should be used for Risk Management and insurance placement purposes only.

The visit comprised an inspection of specific areas concerned of the site, except tenanted areas, F&B tenants such as kitchen, etc., were not accessible at time of the visit.

Our visits will use National Fire Protection Association (NFPA), FM Global and local guidelines as technical references. Each site visit included the following activity:

- (a) Physical tours of [PROPERTY] T3 occupied spaces to observe occupancy related property hazards, storage practices and identify areas not in accordance with the above guidelines.
- (b) A review of existing loss prevention programs, procedures, reports, and documentation to verify status in accordance with corporate standards and practices.
- (c) Interview staff to gain a perspective on loss prevention programs and implementation to understand issues facing local management.
- (d) An exit meeting with site management to achieve agreement with our conclusions and to develop a recommended action plan.

This survey also included a review of administrative and procedural controls such as inspection, testing, and maintenance of fire protection systems and equipment; impairments to these systems; and fire emergency planning.

No physical tests were witnessed during the survey.

Section 2

Executive Summary

The [PROPERTY] ([PROPERTY]) is a mixed-use development that occupies a [AREA] waterfront site at the heart of Singapore's New Downtown. The entire development comprises of 3 office towers ([PROPERTY]s 1, 2 and 3), 2 residential towers ([LOCATION] Residences and [LOCATION] Suites) and a retail mall ([LOCATION] Link Mall). [PROPERTY] was constructed in 2 phases, with its first phase completed in 3Q 2010 and its second phase in 2012.

The first phase consists of [PROPERTY]s 1 and 2, [LOCATION] Residences and [AREA]m² of the retail mall. The second phase comprises [PROPERTY], having 7,850m² of the retail mall and [LOCATION] Suites. The scope of this report is limited to [PROPERTY]s 3, a 46 storey office building with 3 basement levels. [PROPERTY] is jointly owned by [PROPERTY OWNERS] through an investment holding company, [PROPERTY] LLP. [CLIENT] is the asset and facilities manager for [PROPERTY] [PROPERTY].

[PROPERTY] is a 99-year leasehold property (commencing from [DATE]) which obtained its TOP in [DATE]. It is of fire-resistive construction with reinforced concrete load-bearing columns/beams, reinforced concrete floor slabs and a concrete flat roof. Internal walls are of masonry construction except for the reinforced concrete core structure. The building features a 3 storey retail podium at its base which houses a number of restaurants, a food court, in addition to a banking hall.

Sprinkler protection is provided throughout all buildings within the [PROPERTY] development with exception of the electrical equipment rooms and riser compartments (provided with spot-type smoke detectors). The sprinkler system is designed in accordance to CP52: 2004 (Code of practice for automatic sprinkler system). Automatic fire detection in the form of spot-type smoke detectors is provided within all electrical equipment rooms and electrical / telecommunication riser compartments throughout the building. Smoke detectors have also been installed throughout the 3 storeys of the retail podium, while heat detectors have been installed within individual outdoor generator set enclosures.

Additional fire protection equipment comprise of an engineered smoke control system in the podium, in addition wet risers, internal hose reels, portable fire extinguishers and manual break glass call points installed at strategic locations throughout the development. The buildings are equipped with a fire alarm system which is monitored 24/7 both locally and centrally via a DECAM station, Chubb.

The [PROPERTY] development is overseen by a property management team from [CLIENT]. The management team has established a high standard of loss prevention programmes for the site: 24/7 onsite security provided via contract guards; documented emergency response plan and trained emergency response team; formal permit-to-work system and hot work permit programme; good housekeeping practices; and a comprehensive preventive maintenance program for mechanical, electrical and fire protection equipment, among others.

Natural catastrophe exposures for the site are generally low, with exception of flood (moderate exposure) and lightning (moderate to high exposure).

Changes since Previous Survey

1. Added 3 nos of EV charging lots at B1.
2. Upgraded the building fire alarm system, software and some hardware upgrade.
3. Upgraded the building CCTVs to fully digital with video analytic capability.

Planned Changes / Projects

Ongoing AHU chilled water control valves replacement to smart energy control valves. Project expected to be completed by end 2026.

Section 3

Opportunities for Improvement

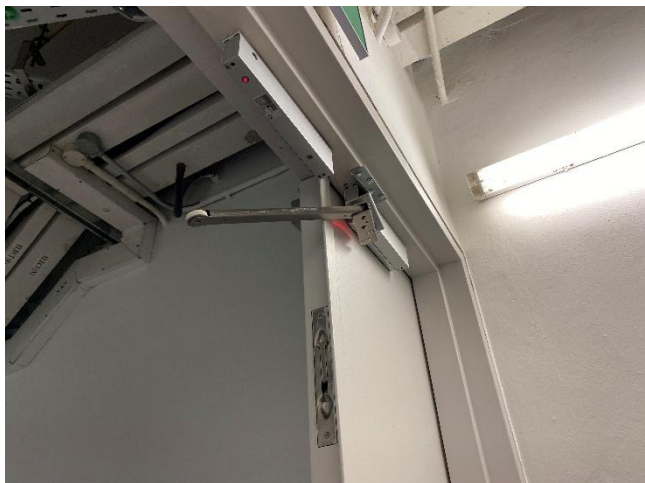
Opportunities for Improvement (OFI) are suggested to mitigate risks. They are intended to create value and manage risk exposures while maintaining relevance to your business objectives. The Opportunities for Improvement are suggested as general sound risk management to implement at the location.

Prioritization of OFI's are classified in line with the following indicative Risk Assessment Matrix (RAM). The relative risk associated with each OFI is measures the estimated consequences of the damage versus the likelihood of occurrence.

Risk Assessment Matrix (RAM):

Consequences					
Likelihood	Insignificant	Minor	Moderate	Major	Catastrophic
Almost Certain	Priority 2	Priority 2	Priority 1	Priority 1	Priority 1
Likely	Priority 3	Priority 2	Priority 2	Priority 1	Priority 1
Possible	Priority 4	Priority 3	Priority 2	Priority 1	Priority 1
Unlikely	Priority 4	Priority 4	Priority 3	Priority 2	Priority 1
Very Rare	Priority 4	Priority 4	Priority 3	Priority 2	Priority 2

New OFI

2026-01		Fire door inspection and testing	
Status	Open	Issue Date	26-February-2026
Priority	Priority 1	Type	Fire door ITM
OFI Issued By	Marsh		
Related RTM ID	N.A		
Description	Incorporate fire door inspection and testing into the maintenance regimen. Per NFPA 101, fire doors and its assemblies should be inspected annually and kept in always working conditions.		
Observation	Inspection and testing of the fire door is not incorporated into the maintenance regimen. As observed at B2 service lift corridor, fire door sequencer is not in good working condition. Currently, inspection of the fire door is not incorporated into the maintenance schedule.  Door sequencer not in good working condition		
Client Response	Client will implement the OFI accordingly.		
Effort of Implementation	Moderate procedural		

2026-02		Sprinkler Installation	
Status	Open	Issue Date	26-February-2026
Priority	Priority 1	Type	Sprinkler Installation
OFI Issued By	Marsh		
Related RTM ID	N.A		
Description	Ensure that sprinklers are properly installed and not recess into the ceiling.		
Observation	As observed at B3 service corridor area, a few of the sprinkler heads are recessing more than 50% into the ceiling. This will impede the flow pattern of the sprinklers.		
			
	Sprinkler head recess into ceiling		
Client Response	Client will implement the OFI accordingly.		
Effort of Implementation	Moderate procedural		

Update of Previous Recommendations

Rec No.	Description	Status
2022-01	Consider BCP for Critical Utility Failure	Completed. Critical utility failure is part of the BCP scenarios.
2022-02	Review Location of Fire Extinguisher	Completed. Location reviewed accordingly.
2022-03	Ensure Proper Sealing of Utility Penetration Thru Fire Compartment	Completed.
MRC CBD-20-01	Establish fire protection system impairment plan	Completed.
MRC CBD-20-02	Establish portable electrical appliance inspection and testing program	Completed. Based on inspection record submit on 11 Feb 22, quarterly inspection for portable electrical appliance is in place.
MRC CBD-20-03	Provide thermal runaway monitoring for VRLA batteries	Completed. Per email received on 13 March 2020, [CLIENT] do conduct regular maintenance and servicing and cyclical batteries replacement with the UPS fault alarm connected to Landlord BMS system.
MRC CBD-20-04	Replace kitchen fire suppression discharge nozzle end caps at NTUC Foodfare food court	Completed.

Section 4

Loss Estimates

The Loss Estimates presented here are believed to be reasonable, based on industry experience, events postulated, and information provided by the client. The calculation of Loss Expectancies is based on the review of building construction, operations, fire protection systems, and fire protection features at the time of our assessment. The estimates are further based on conditions observed at the time of the visit. By their nature, these estimates contain some element of subjectivity. Accordingly, the estimates cannot be taken as absolutes and could be exceeded due to changes in physical conditions on site, or the initiating event or escalation being more severe than anticipated within the boundaries of the estimate.

All damage and loss potential figures presented pertain exclusively to primary property damage, associated contents damage, and associated business interruption recovery time loss, caused directly by (fire or explosion) as defined in our Loss Estimate.

Values

Date of values	2026 / 2027	Currency	SGD
Property Damage (PD)		Business Interruption (BI)	
[PROPERTY]	[VALUE]	Gross revenue (based on 46 months indemnity period)	[VALUE]
Demolition and Debris Removal	[VALUE]	Additional increased cost of working	[VALUE]
Professional Fees at 7%	[VALUE]		
Site Inspectorate	[VALUE]		
Authorities' fees and charges	[VALUE]		
Allowance for Inflation	[VALUE]		
FM Office, M&E Office, Training Room & Building Supervisor Office	[VALUE]		
Inventory Stock at L2, L39 & B2	[VALUE]		
EV Charging Points	[VALUE]		
Total Site PD value	[VALUE]	Total Site BI value	[VALUE]

The PD values on building and infrastructure were provided by [CLIENT] and should be based on Replacement Cost Values (RCV).

Loss Estimates Definitions:

Level	Definition
Level I	<u>Primary Protection Systems are functioning and manual firefighting available</u> A loss event in which damage is based on the nature of hazards and construction factors, and where: • All fire protection systems are in service and functioning as designed. • Full facility Industrial Fire Brigade (Or Plant Emergency Organization) and Municipal Fire Department response expected. • Credit is given to all properly maintained fire barriers up to their design duration rating • Construction features function as designed.
Level II	<u>Primary Protection Systems are not functioning, Manual firefighting response only</u> A level II Loss Event is one which occurs when: • The fire protection system protection the area with the largest PD / BI potential is impaired or is rendered inoperative or ineffective due to the nature of the event. Adjacent fire protection systems are presumed operational unless rendered inoperative or ineffective due to structural failure. Same applies for the use of special extinguishing systems. • Credit can be given for adequate manual emergency response • Credit given to minimum adequately maintained (including fire doors and fire penetrations) 3 hour rated walls where the combustible loading is light to ordinary, structural failure is not expected, and roof assembly is a listed or approved noncombustible. • Combustible roof construction (including combustible or unknown metal deck assemblies) results in a contiguous structure loss. Damage may be limited to the area where the impaired protection system is located and the nearby surroundings or may extend to the nearest adequate separation or properly designed and approved construction cutoffs, depending on site conditions.
Level III	<u>Fire Protection Systems are not functioning and no manual firefighting response</u> A level III Loss Event is one which occurs when: • All fire protection systems throughout the entire site or facility are impaired. • No credit is given for manual emergency response. • Damage is limited only by adequate separation and/or free-standing 4-hours rated firewalls or equivalent. (Equivalencies must be well defined and proven.) • Combustible roof construction (including combustible or unknown metal deck assemblies) results in a contiguous structure loss. The size of this loss can approach the value of the buildings of origin or an entire facility, depending on site layout.

Property - Level II Loss Estimate

Probable Maximum Loss (PML)

Definition	Definition: Primary Protection Systems are not functioning, Manual firefighting response only
Scenario	The PML scenario will be a fire starting within a tenant space on Level 27 with internal staircase leading upto L29 of [PROPERTY]. The fire may be caused by one of the following: fault in electrical fixture/appliance, improperly managed hot work activity, smoking, etc. As the building's sprinkler and automatic fire detection systems are assumed to be impaired (per the PML definition) the fire will reach a fully grown state and spread across the L27 floor plate due to an absence of horizontal fire compartmentation, and via continuity of combustibles. The fire will also spread vertically to 2 upper floors via internal unenclosed staircases from L27 to L29 (fire shutters won't be operational as fire detection systems are assumed to be impaired), prior to response by the emergency response team/fire brigade. The fire is expected to be contained at L29 with the timely response of emergency responders. Smoke, fire and water damage is anticipated to be sustained on L30 to L31, in addition to water damage in varying degrees to 5 floors below L27 (L22 to L26).
Assumptions	Each floor from L27 to L29 is assumed to be 100% damaged (from fire/smoke), L30 to L31 is assumed to be 30% damaged (from smoke/water), while each floor from L22 to L26 is assumed to be 15% damaged (from water). Total number of floors in [PROPERTY] = 46 + 3 Estimated cost per floor of [PROPERTY] 3 = [VALUE] / 49 = \$[VALUE].

Property Damage	Insured Value (SGD)	% Damage	Property Damage
L28 to L30	3 x [VALUE]	100%	[VALUE]
L30 to L31	2 x [VALUE]	30%	[VALUE]
L23 to L26	5 x [VALUE]	15%	[VALUE]
Total Property Damage Loss			\$ [VALUE]

Business Interruption

It will take about 1 month for investigation and clean-up, up to 7 months for full reinstatement works involving the fire-damaged floors (L27 to L31) to be completed. Tenants will not be able to operate out of the building/affected floors during the clean-up/reinstatement/recovery period and will have to relocate to other sites. For L22 to L 26, tenants are still able to operate out of the existing units with ongoing repair for water damages.

Assuming lost rental income for an 8 month period, the BI PML is estimated (based on a percentage of affected lettable area) as follows:

Total lettable area is [AREA] m²

Typical floor net-lettable area is estimated to be 2,800 m².

Total affected floor = 5 x 2,800 = 14,000 m²

BI PML = [VALUE] × 8/46 × [VALUE] = \$[VALUE]

Increase cost to working = \$[VALUE]

Total Business Interruption Loss

\$[VALUE]

Total Level II PD/BI Loss Estimate

\$[VALUE]

Property - Level III Loss Estimate

Maximum Foreseeable Loss (MFL)

Definition	Definition: Fire Protection Systems are not functioning and no manual firefighting response
Scenario	The MFL scenario will be a fire starting within a tenant space on Level 27 (with internal staircase upto L29) of [PROPERTY]. The fire may be caused by one of the following: fault in electrical fixture/appliance, improperly managed hot work activity, smoking, etc. As the building's sprinkler and automatic fire detection systems are assumed to be impaired the fire will reach a fully grown state and spread across the L27 floor plate due to an absence of horizontal fire compartmentation, and via continuity of combustibles. With the local fire brigade assumed to be delayed for 2 hours (as per MFL), the fire will spread vertically to the upper floors via internal unenclosed staircases, breached floor penetrations and the building's exterior glass curtain walls, resulting in fire and associated smoke damage to 10 upper floors (i.e. L27 to L36). Water damage from a manual fighting (after 2 hours) will result in water damages in varying degrees to 10 floors below L27 (L26 to L17). The fire is not anticipated to spread to other Towers due to spatial separation and lack of continuous combustibles.
Assumptions	Each floor from L27 to L36 is assumed to be 100% damaged (from fire/smoke), while each floor from L26 to L17 is assumed to be 15% damaged (from water). Total number of floors in [PROPERTY] 3: 46 + 3 Estimated cost per floor of [PROPERTY]: [VALUE] / 49 = \$[VALUE].

Property Damage	Insured Value (SGD)	% Damage	Property Damage
L27 to L36	10 x [VALUE]	100%	[VALUE]
L17 to L26	10 x [VALUE]	15%	[VALUE]
Total Property Damage Loss*			[VALUE]

*Excluding ancillary charges for demolition and debris removal, professional fees, etc.

Business Interruption

It will take about 2 month for investigation and clean-up, up to 14 months for full reinstatement works involving the fire-damaged floors (L27 to L36) to be completed with another in addition to another 2 months for recovery of business to previous levels. Tenants will not be able to operate out of the building/affected floors during the clean-up/reinstatement/recovery period and will have to relocate to other sites.

For L26 to L17, tenants are still able to operate out of the existing units with ongoing repair for water damages.

Assuming lost rental income for an 18 month period, the BI PML is estimated (based on a percentage of affected lettable area) as follows:

Floor plate area of typical level (L27 to L36) in [PROPERTY] = 2,800m²

Total affected floor area = 10 × 2,000 = 28,000m²

Total net lettable area of [PROPERTY] ~ 125,400m²

BI PML = 28,000/125,400 × 18/46 × [VALUE] = \$[VALUE] + \$[VALUE] (increase cost of working)

Total Business Interruption Loss	\$[VALUE]
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Total Level III PD/BI Loss Estimate	\$ [VALUE]
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Section 5

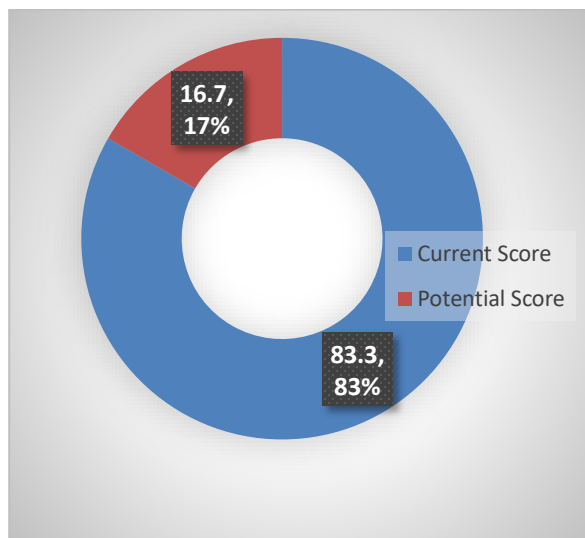
Risk Quality Ratings and Comments

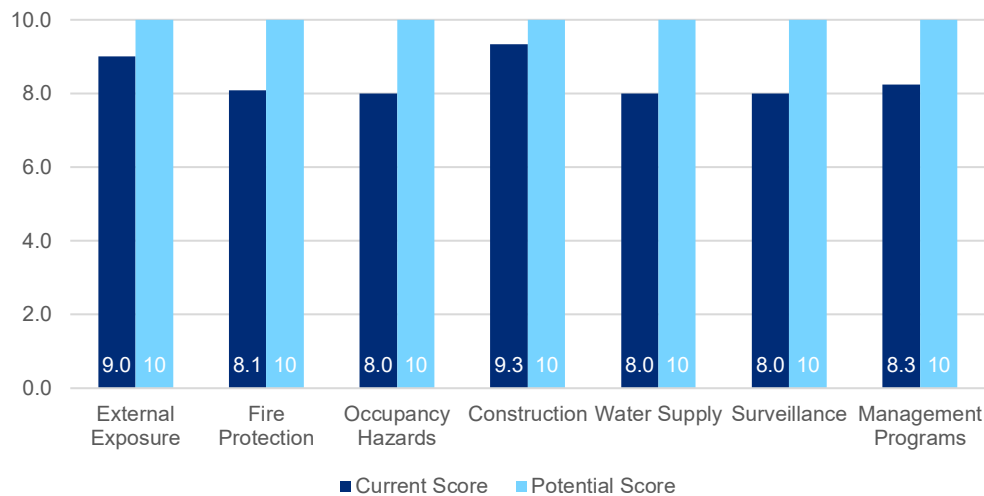
Our site visit, interviews and review of information provided lead us to the following maturity assessment.

Main Category	Current Score	Potential Score
External Exposure	9.0	10
Fire Protection	8.1	10
Occupancy Hazards	8.0	10
Construction	9.3	10
Water Supply	8.0	10
Surveillance	8.0	10
Management Programs	8.3	10
Location Risk Quality Score	83.3	100.0
Location Risk Quality Rating	B	A

Rating Legend
A: 85 to 100
B: 70 to 84
C: 51 to 69
D: 36 to 50
E: 0 to 35

Current Risk Quality Score versus % Potential



Risk Quality Rating Category Summary

Overall risk rating for the site can be considered to be above average risk of its class. The property embodies some of the best practices in the industry and acceptable standards are exhibited.

The positive points which come out from the visit are following:

- Buildings were constructed of fire resistive construction materials.
- The buildings are fully sprinklered except for electrical rooms and electrical / utility risers which are provided with smoke detectors.
- Trainings for employees regarding fire life safety awareness at the workplace, fire prevention and action to take should a fire occur.
- Preventive maintenance programs are in place for critical utilities and fire protection and detection equipment.
- Good redundancy for critical utilities.
- Good housekeeping observed for all areas visited.

Section 6

Management Programs

Our site visit, interviews and review of information provided leads us to the following maturity assessment of Management Programs in place to manage the risks at the location.

Program Category	Maturity	Comments
Building Maintenance	Embedded	Regular preventive and corrective maintenance is carried out for the buildings/finishes. Structural inspection of the buildings is conducted once every 5 years by appointed professional civil engineer with report submitted to the local authority, BCA. The next structural inspection is expected to be conducted in 2027. A preventive maintenance (PM) schedule incorporating inspections and equipment servicing/testing has been established for building utility and fire protection equipment. A computerized maintenance management system (Build Folio) is in use for the monitoring/processing of work orders and upcoming maintenance activities.
Electrical Maintenance	Embedded	Planned maintenance programme is carried out by electrical contractor. Inspection, testing and maintenance are conducted by third party electrical contractors and Licensed Electrical Workers (LEWs). Maintenance activity includes, but not limited to the following: • Annual shutdown maintenance for HT/LT switchgears. • Partial discharge on switchgears conducted annually for MV and HT. • IR Thermography Scanning: inspection conducted yearly by appointed electrical contractor for HT / LT switchboards and to downstream distribution boards. Quarterly scanning conducted by in-house technician. • Generators are tested monthly with partial load (using a load bank), and with their full loads (building) during the annual power shutdown maintenance.
Equipment Maintenance	Embedded	Comprehensive maintenance contract by appointed Integrated Facility Management (IFM), Jardine Engineering Corporation (JEC),

Program Category	Maturity	Comments
		for maintenance of key mechanical, electrical and plumbing equipment.
Fire Protection Impairment Management	Embedded	A permit is utilized for any impairment to the buildings' fire sprinkler and alarm systems. All proposed sprinkler/alarm impairments are required to be approved by [CLIENT].
Fire Protection System Maintenance & Testing	Embedded	A service contract is in place with appointed fire contractor for the preventive maintenance of the site's fire protection & detection equipment/systems. A master testing schedule and plan is available. Maintenance includes a monthly functional testing of fire pumps, an annual sprinkler system water-proving test, annual inspection/servicing of fire extinguishers and testing of at least 20% of the smoke detectors annually among others.
Hazardous Materials Handling	Optimized	There is no hazardous material that is handled or stored.
Housekeeping	Embedded	There is a formal system for the collection and disposal of different type of waste generated in the site. Housekeeping was observed to be of a good standard within visited building common areas and electrical equipment plant rooms, which were noted to be kept clear of storage.
Hot Work Permit System	Embedded	Formal hot work permit system and procedures have been established by [CLIENT]. The hot work permit is subject to sign-off/approval by the FSM, with a building supervisor required to conduct a work site inspection to verify that all necessary safety precautions are taken. Precautions include a continuous fire watch during and for 1 hour following the completion of works. Permits have a validity period of 1 day.
Management of Change	Embedded	Not required in this occupancy.
Pre-Emergency Planning	Embedded	A documented emergency response plan has been established. A company emergency response team (CERT) has been established for the facility. Per shift there are 5 first responders on duty. The CERT comprises of

Program Category	Maturity	Comments
		staff from facility management and security. CERT members are required to undergo basic firefighting/rescue and first aid training by SCDF-certified third party training centres. Monthly in-house training is conducted for the CERT members. A certified Fire Safety Manager (FSM) has also been engaged to maintain and enhance fire safety standards within the development. Fire evacuation drills and tabletop exercises are carried out twice a year.
Self-Inspections	Embedded	There are various self-inspection conducted. Building services will conducted daily zonal inspection. Engineering technicians perform daily visual checks of utility plant rooms and equipment to identify and address anomalies. A safety committee comprising of [CLIENT] staff and vendors has been established which conducts in-house safety inspections once every 2 months. Daily fire safety and general housekeeping inspections of building common areas are conducted by the FSM. During any project fit-out work, daily inspection will be conducted for the fit-out area.
Smoking Controls	Optimized	A “no smoking” policy is in place, with “no smoking” signs posted throughout the compound’s common areas. Smoking is only permitted at a designated outdoor smoking point facing [ROAD]. The smoking point is provided with stainless steel bins for cigarette butt disposal. During the site visit, no causally discarded cigarette butt observed in the areas inspected.
Contractor Management	Embedded	An approved vendor list is in use and this is reviewed annually. A formal permit-to-work system has been established for contractors. A general work permit form is in use, with separate permits issued for hazardous work activities such as hot work and work at height. Contractors have to undergo a safety & health induction training provided by the EHS manager.

Program Category	Maturity	Comments
Business Continuity Planning	Embedded	There is a local level BCP document that is synchronized with the corporate plan for all the 3 entities. There are 18 scenarios formulated and include fire drills. Annual exercises are conducted for the different scenarios with activation of key personnel and call tree exercises.
Environmental, Social & Governance (ESG)	Not Evaluated	

In assessing the maturity of the Management Programs the following maturity scale was applied:

Maturity Rank	Definition
Undeveloped	Informal actions with little or no systematic procedures.
Formalized	Programs and procedures are established. Full familiarity is uncertain.
Established	Programs and procedures are communicated throughout the organization. Orientation is ongoing.
Embedded	Ownership is established at most or all organizational levels. Training and some exercises is conducted.
Optimized	Full ownership with active program maintenance, testing, exercising and continuous improvement.

Management Program Comments

Based on site discussion and observation, [CLIENT] in [PROPERTY] has established good management program / procedures / policy and fulfilling local regulatory requirements.

Section 7

Construction

[Site Map]

Source: One Map

Construction Narrative

[PROPERTY] is a 46-storey building with 3 basement levels. The building is of fire-resistive construction with reinforced concrete load-bearing columns/beams, reinforced concrete floor slabs and a concrete flat roof. Internal walls are of masonry construction except for the reinforced concrete core structure. Partition walls within the office spaces comprise of drywall constructed from gypsum boards, aluminium stiffeners and rock wool insulation in between. The building has a unitized aluminium glass curtain wall facade with double glazed low E-solar tinted or single thermal toughened glass. Aluminium spandrel panels have been fitted between the glass curtain wall panels (level with the building floor plates) and these are bonded with 50mm thick rock wool insulation. The gaps between the curtain wall/spandrel system and concrete floor slabs are filled with 135mm thick rock wool (2 hours fire rated). There is no combustible insulation utilized on the facade of [PROPERTY].

[PROPERTY] has a total gross floor area (excluding car park areas but including car park auxiliary areas) of [AREA]m². The [PROPERTY] car park occupies the 3 basement levels and has a total of 351 parking lots in addition to 6 handicap lots. Car parking areas take up a total of 19,567m², with floor area breakdown (across the 3 basement levels) as follows:

- Basement 1 – [AREA]m²
- Basement 2 – [AREA]m²
- Basement 3 – [AREA]m²

[PROPERTY] was awarded the Building Construction Authority (BCA) Green Mark Platinum Award (re-certified every 3 years) and has the following green building features (among others):

- Low E-coating glass facade
- Extensive use of LED lightings with motion sensors and photocell sensors
- Use of District Cooling System to provide efficient air-conditioning
- Heat recovery wheel (energy recovery heat exchanger)
- Lifts with regenerative drive system
- Greenery provision at L1 and L5
- High efficiency filters for air handling units (AHUs)

Summary of construction as follow:

Construction Table

Building	Year built	Stories	Area (m.sq)*	Roof Const.	External Walls	Floor Const.	Column	Est. Height (m)	Comb / Non Comb	% Sprinkl ered	% Sprinkl ers Needed
[PROPERTY] 1	2010	B3 + 46	[AREA]	RC	AC	RC	RC	210	Non Comb	100*	NA*

S – Steel; M – Masonry; AC – Aluminium curtain wall facades with double glazed glass, RC – Reinforced Concrete

+ Exclude parking spaces

* Sprinkler protection is provided in all areas except electrical rooms and rooms with sensitive equipment.

Compartmentalization and Fire Divisions

[PROPERTY] and [PROPERTY] is connected at basement 2 carpark area. There is a 2 hours rated fire shutter installed at the interconnection area. The connection is not in use, and the fire shutter is permanently shut.

The building's office floor plates adopt an open-plan concept with no horizontal fire separation within floors (with exception of utility rooms/compartments). Electrical equipment rooms throughout the development are constructed of compartment walls and floors having a fire resistance of at least 2 hours, with fire doors leading into electrical rooms rated to 2 hours. Cable penetrations were noted to be adequately sealed in all visited electrical equipment rooms.

There is vertical communication between B3 and B1 of [PROPERTY] via car parking ramps, and between L1 to L3 via the podium atrium. Electrical and telecommunication riser compartments within the development are provided with 1-hour fire-rated doors. Telecommunication risers are triple volume (1=3 storey) compartments and fire-stopped on every 3 floors. Visited riser compartments were noted to be adequately fire-stopped. There is a combination of 1 hour and 2 × 1-hour fire doors (via smoke stop/firefighting lift lobbies) leading into the individual fire stairwell enclosures within the building.

Knock out panels have been provided on the individual office floor plates for the construction of internal (non-enclosed) staircases within tenant spaces. Currently, there are internal staircases connecting the following floors: L27 to L29, L34 to L35, L40 to L41 and L45 to L46. Smoke detector-actuated 2-hour fire-rated roller shutters have been installed around all internal staircase enclosures at the above locations, serving as vertical fire compartmentation between floors.

Details on Combustible Construction

Based on site observation and information provided there are no combustible material used for construction.

Section 8

Occupancy, Hazards, and Utilities

Occupancy

[PROPERTY] is a 46-storey building with 3 basement levels.

Overview of the key occupancy of the different blocks as follows:

Floor	Occupancy
[PROPERTY]	
B3	Car park, fire pump room, heat exchanger room, chilled water pump room, domestic pump room, lift motor room
B2	Car park, loading/unloading bay, fuel pump room, building management office, engineering workshop and storage rooms
B1	Car park
L1	Main lobby, fire command centre (FCC), restaurants and cafes
L2	Kopitiam food court and [TENANT]'s cafeteria
L3	[TENANT] banking hall, auditorium
L5 to L6	Offices
L7	M&E floor: transformer rooms, high voltage (HV) switch rooms (separate rooms for landlord and tenants), low tension (LT) switch rooms (separate rooms for landlord and tenants), outdoor generator sets, sprinkler control valves, contractor's workshop
L8 to L38	Offices
L39	M&E floor: transformer rooms, HV switch rooms (separate rooms for landlord and tenants) LT switch rooms (separate rooms for landlord and tenants), fire pump room, heat exchanger room, generator room
L40 to L46	Offices
Roof (L47 to L48)	Outdoor generator sets, air-cooled chillers, lift motor rooms

Key Features / Processes

Activity / Processes	[PROPERTY] is occupied primarily as a multi-tenanted office building with a retail podium from L1 to L3, in addition to 3 basement levels of car parking.
Employees / Shifts	[PROPERTY] is managed by a property management team from [CLIENT]. The [CLIENT] team is based in [PROPERTY] and comprise of the following full-time staff: 1 building manager; 2 building executives; 1 technical manager; 6 technical officers; 12 building supervisors; 2 engineers; 3 concierge staff (shared with 2 other entities); 1 environmental, health and safety (EHS) manager and 1 customer service manager. A number of these staff (e.g. EHS manager and customer service manager) have oversight over both the ORQ and [PROPERTY] developments. Official [CLIENT] office hours are from 8.30am to 6pm, 5 days a week. Building supervisors operate 24/7 over 2 shifts with 3 building supervisors' onsite at any 1 time, while technical officers operate from 7.30am to 11pm daily over 2 shifts, with 3 duty officers per shift. Jardine Engineering Company (JEC) has been engaged by [CLIENT] to be the main building/utility maintenance contractor for [PROPERTY]. JEC technicians are based onsite. There is a centralized Building Management System (BMS) monitoring located at [PROPERTY] T2, B2 loading bay. The centralized BMS system monitors [PROPERTY] T1 to T3 and ORQ. JEC manages the properties as a cluster with shared resources. There are 2 engineers and 2 BMS operators and they support all properties. For [PROPERTY] T3, JEC deployed 2 – 3 dedicated technicians per shift.
Third parties/tenanted areas	The occupancy rate is 99%. The building's anchor tenant is [TENANT] which occupies 15 floors (L3, L5, L6, L8 to L12, L40 to L46). Other major tenants include Wong Partnership (occupying 3 floors; L27 to L29) and Rio Tinto (occupying 1.75 floors; L20 to L21). The retail podium is occupied by restaurants/cafes on L1 (ground level); an food court and cafeteria (by [TENANT]) on L2; and the [TENANT] banking hall and auditorium on L3.
Common Hazards	Common property hazards include fire arising from faults in electrical equipment/lighting, improperly managed hot work activities, or commercial cooking operations; and water damage arising from wet pipe sprinkler actuation or chilled/domestic water pipe leaks/bursts.

Special Hazards	There is no bulk storage of flammable liquids or gases within the development. Town gas is supplied to the food court and the restaurants within the podium. This is piped directly into the respective tenant units from the gas supplier, City Gas Pte Ltd. The incoming gas supply arrangements at the individual tenant units are provided with solenoid safety cut-off valves activated by gas leakage detectors within the kitchen areas. Solenoid bypass valves are padlocked (in the closed position) with keys held by building management to ensure that there is no unauthorized opening of the bypass valves in the event of a gas leak. The gas detectors in the respective kitchens are not linked to the BMS or FCC. The main incoming gas supply arrangement for the development is not provided with a solenoid safety cut-off valve. There are however, gas leakage detectors provided at the gas supply valve chamber, with alarm(s) monitored at the FCC and BMS. Kitchen exhaust ducts extend from the individual tenant kitchens and combine into a central exhaust duct which terminates on L5 and L7. The responsibility for cleaning of kitchen exhaust hoods and ducting extending from the F&B kitchens lies with the respective tenants; this is spelled out in [CLIENT]'s management regulations. Quarterly cleaning is conducted by specialist contractor engaged by the tenants. Service reports verifying the cleaning of kitchen ventilation/ exhaust systems are obtained by [CLIENT] from the tenants. The development's central exhaust ducts (responsibility of [CLIENT]) are cleaned on an annual basis by a third party contractor.
Storage	Limited to small pockets of back-of house storage belonging to retail tenants. Engineering equipment/raw materials/spares storage located in B2 storage rooms control by [CLIENT]. There are no major warehousing facilities in the development.

Utilities and Infrastructure

Power

Distribution on Site	Underground cables.
Details	2 underground incoming feeders supplied at 22kV. The incoming power is stepped down before being distributed to the respective electrical rooms and distribution boards.
Back-up Power	Other than the normal power supply from the grid, emergency power diesel engine generators are provided to maintain essential and life safety equipment within the development. Back-up power is provided for building essential and emergency services only by 5 outdoor landlord-owned diesel generator sets: - 2 x 1.75MVA at L7 - 1 x 1.5MVA at L39 & - 2 x 2MVA at building roof In addition, tenant owned ([TENANT]) generators as follows: - 5 x 1.5MVA at L7 - 2 x 1.5 MVA at building roof

Transformer Details

There are 13 landlord-owned transformers supplying to [PROPERTY], details as below:

Transformer location	Number	Rating (MVA)	Make/Type	Serving
L7 transformer rooms	6	2.5 each	ABB, Cast resin dry type	4 – landlord areas; 2 – tenant areas
L39 transformer rooms	7	2.5 each		4 – landlord areas; 3 – tenant areas

There are other tenant-owned transformers installed in separate rooms on L7 and L39. All landlord-owned transformer units are indicated to be running (none on standby) with average utilization rates of less than 50% of their designed capacities. A manual coupling (estimated 2 hours lead time) is required to be performed by the appointed licensed electrical worker (LEW) for switching of building load between transformer units.

Transformer rooms (each housing 1 to 2 transformers) are a minimum of 2 hour fire compartments with 2 hour fire doors, however there are no fire walls separating transformer units within each room.

Boilers

None installed.

Chillers / Cooling Towers

Chilled water for the building's central air-conditioning system is supplied via a district cooling system (DCS) operated by Singapore District Cooling (located at 2 sites: B2 of ORQ and at the [LOCATION] Sands).

The development's air-conditioning system comprises of a network of air-handling units (AHUs) and heat exchangers, supported by a chilled water circulation system driven by a number of circulation pumps. There are 4 x heat exchangers and 5 x chilled water pumps in B3. This is in addition to a secondary chilled water circuit with 3 x heat exchangers and 3 x circulation pumps on L39. The B3 heat exchangers are SDC-owned while the L39 heat exchangers are landlord-owned. All chilled water pumps are landlord-owned and installed with at least N+1 redundancy. Each level of [PROPERTY] is served by 2 AHUs with ducted return.

In the event when chilled water supply from DCS is disrupted, there are 5 x 400RT landlord-owned air-cooled chillers (all on standby) on the building roof which serve as backup chilled water supply for essential utility rooms, as well as for tenants' use (by agreement only). These chillers are backed up by emergency power and can support up to 20% of the buildings' total cooling requirements. [TENANT] has also installed it's own chillers (on L7) for backup chilled water supply.

Landlord-owned air-conditioning systems (AHUs, pumps, air-cooled chillers, VRV units, etc.) are serviced and maintained on a monthly basis by JEC.

There are no cooling towers installed.

Fuels and Energy Supplies

Each generator set is installed within a metal enclosure and is supplied with a ~700L diesel day tank. Day tanks are double-walled with leak detection provided between the inner and outer walls; these are monitored via the BMS. The day tanks are supplied via 3 x 30,000L underground storage tanks (USTs) located in B3. Diesel from the USTs is automatically pumped to the generator day tanks via transfer pumps (in the B2 fuel pump room) when a pre-set low level is reached.

The generator enclosures are provided with spot-type heat detectors linked to solenoid shut-off valves on the incoming diesel pipelines; these will cut off diesel supply in the event of a fire within the enclosure. The 3 USTs are installed within individual 4-hour rated sand-filled concrete enclosures which are provided with dedicated leakage detection sensors.

Section 9

Fire Protection

Site Fire Protection Abstract

The fire protection water supply system, automatic water-based fire protection systems, fire detection & alarm systems, and manual fire-fighting means installed within the site are primarily designed and installed to local code, namely: CP 52:2004 (Code of practice for automatic fire sprinkler system), CP 10:2005 (Code of practice for the installation and servicing of electrical fire alarm systems) and SS 546:2009 (Code of practice for emergency voice communication systems in buildings). The major components of these systems are UL listed and/or FM approved.

Engineered smoke control system (ESCS) – Installed in the podium. The ESCS on L1 was designed by Arup and is a performance-based system, while that on L2 and L3 is designed by Meinhardt and is a prescriptive-based system (per the local fire code). The ESCS comprises of a network of smoke curtains, smoke dampers and smoke extraction fans, controlled and synchronized by an addressable smoke detection system. The podium is segregated into multiple smoke control zones, with dedicated smoke dampers, extraction fans and detectors for each zone. Smoke is exhausted via a common duct (which traverses through all zones), with exhaust vents located on L5 and L7.

The engineered smoke control system is serviced and maintained every month by JEC. Smoke curtains and exhaust fans are tested at least twice a year.

Automatic Sprinkler Protection

Automatic sprinklers are installed as follows.

Coverage	Entire development with exception of electrical equipment rooms and riser compartments which are provided with smoke detectors.
Design standard	CP 52: 2004 (Code of practice for automatic fire sprinkler system): combination of OH1 (5 mm/min @ 72 m ²), OH2 (5 mm/min @ 144 m ²) and OH3 (5 mm/min @ 216 m ²).
Sprinkler System Monitoring	The sprinkler system is served by a total of 30 sprinkler control valves (SCVs) located on various floors of the building; there is 1 SCV for every floor from B3 to L8, and 1 SCV for every 2 floors from L9 to L48 (roof). SCVs are strap locked (in the open positions) to mitigate against inadvertent/malicious tampering and are monitored with tamper switches.

System Maintenance	The sprinkler systems are preventively maintained by the appointed fire protection system maintenance contractor. This includes monthly cut-in/cut-out testing of the fire pumps, in addition to inspections of system components and indicator panels. An annual water proving test is carried out at the individual SCVs as part of fire certificate renewal. Refer to Appendix for the most recent sprinkler system water-proving test results.
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Special Extinguishing Systems

Wet chemical suppression systems have been installed under the kitchen hoods of F&B tenants (with commercial cooking operations) throughout the development. Systems are of different make, with those in visited kitchens noted to be UL 300 listed. Responsibility for system maintenance lies with the respective tenants; this is spelled out in [CLIENT]'s management regulations.

A number of building tenants have installed their own total flooding gaseous suppression systems (e.g. FM 200) to protect their high-asset IT equipment/server rooms. Responsibility for system maintenance lies with the respective tenants.

Water Supply Details

The sprinkler system for [PROPERTY] is designed in accordance to CP52: 2004 for OH 1 and 3; OH3 system covers the basement and podium levels, while OH1 system covers all other areas.

The sprinkler system is supplied via a combination of gravity feed and 4 sets of sprinkler booster pumps, each set comprising of 1 duty pump, 1 standby pump and 1 jockey pump drawing suction from dedicated fire water tanks. All pumps are driven by electric motors and provided with backup power supply (via diesel generator sets). Details on the pump systems are provided below:

Sprinkler Pumps	Location	Areas Served	Make/Model/Type	Nominal Rating/ Hazard Group	Fire Water Tank
SP 1-1	B3 fire pump room	B3 to L3	Ajax-Elite / 80-20 AXOC ZF / Electric	OH3 – 30m head	399m ³ reinforced concrete tank; shared with wet riser and transfer pump systems
SP 1-2			Electric	-	
SJP 1		L5 to L12	Ajax-Elite / E65-20 A20C / Electric	OH1 – 75m head	
SP 2-1				-	
SP 2-2			Electric	-	
SJP 2					
SP 3-1	L39 fire pump room	L27 to L38	Ajax-Elite / 50-16 AXOC ZF / Electric	OH1 – 75m head	188.99m ³ reinforced concrete tank; shared with wet riser and hose reel systems
SP 3-2				-	
SJP 3			Electric	-	
SP 4-1		L39 to L48	Ajax-Elite / E65-20 A20C / Electric	OH1 – 75m head	
SP 4-2				-	
SJP 4			Electric	-	

Sprinkler coverage from L13 to L26 of [PROPERTY] is gravity-fed from the L39 fire water tank. The fire water tanks are supplied via the public mains and also supply to the building's wet riser (both tanks), transfer pump (B3 tank) and hose reel (L39 tank) systems. There are a set of transfer pumps (1 duty, 1 standby; both Ajax Elite MT100/3 – A20C pumps with rating of 12.5L/2 @ 215m head) in the B3 fire pump room which supply water from the B3 tank to the L39 tank.

Pump Flow Test Data

Not Conducted.

Manual Fire Fighting Features

Distance to Fire Department	< 1.5 km to [LOCATION] Fire Station
Fire Department Type	Public fire brigades by Singapore Civil Defence Force (SCDF)
Response Time (minutes)	< 10 minutes
Hydrant Coverage	There are 3 private yard hydrants strategically provided connected directly to public water.
Fire Hydrant Type	Below ground, private, with double pillar outlets.
Industrial Fire Brigade	NA
Portable Extinguishers	Portable fire extinguishers have been provided at strategic locations throughout the development. These comprise of ABC dry powder type extinguishers of various sizes. Fire extinguishers within building common areas are inspected on a monthly basis by in-house technicians and serviced/maintained on an annual basis by contractor. Extinguishers were noted to be adequately tagged with their service records and clearly marked with signage.
Hose Stations / Reels	Full coverage per local codes. Internal 25mm diameter, 30m length hose reels have been provided at strategic locations throughout the common areas of the development. There are 5 hose reel stacks extending throughout all floors of the tower. The hose reel system is supplied via a combination of gravity feed and a total of 2 sets of hose reel booster pumps, each set comprising of 1 duty and 1 standby electric motor-driven pumps. The pump sets are installed in the B3 fire pump room (serving B3 to L9) and on the building roof (serving L38 to L48), with system coverage for all other floors being gravity-fed from the L39 and roof fire water tanks. Hose reels systems are serviced and tested on an annual basis, with visual inspections of individual hose reels carried out on a half-yearly basis.
Risers (Dry / Wet)	[PROPERTY] is protected by a wet riser system, with 2 wet riser stacks extending throughout all floors of the tower, and an additional 5 stacks covering the podium section. The system is supplied via a combination of gravity feed and 2 sets of wet riser booster pumps, each set comprising of 1 duty, 1 standby and 1 jockey electric motor-driven pumps drawing suction from dedicated fire water tanks.

Other Manual Firefighting Appurtenances / Comments:

3 street hydrants strategically provided connected directly to public water. Public street hydrants are drawing water directly from the municipal water supply which is considered as very reliable in Singapore.

Fire Alarm Monitoring

Automatic fire detection in the form of spot-type smoke detectors is provided within electrical equipment rooms (transformer rooms, electrical switch rooms, MDF rooms, etc.) and electrical/ telecommunication riser compartments throughout the building. Smoke detectors have also been installed throughout the 3 storeys of podium. Heat detectors have been installed within the individual outdoor generator set enclosures.

Manual break glass call points with fire alarm bells are installed at strategic locations throughout the development.

The fire alarm system and fire detectors are serviced and tested by Siemens monthly.

Alarm Signaling Services	The main fire alarm panel is located within the FCC on L1 which is manned 24/7. The fire alarm signal is monitored by appointed third party vendor, Chubb, known as providing Decentralised Fire Alarm Monitoring (DECAM) services. Usually within a timeframe of two minutes or shorter, the local fire brigade will be notified via DECAM services if there is no response by the local site on any fire alarm signal received.			
Fire Panel Model / Manufacturer	Siemens			
Alarm Signals Monitored	☒ Water-flow	☒ Valve Tamper	☐ Building Temp	☐ DPV Supervisory
	☒ Smoke Detection	☒ Pull Stations	☐ Heat Detection	☒ Special Ext. Sys.
	☒ Pump Running	☒ Pump Trouble	☒ Tank Level	☐ Tank Temperature
	☐ Intrusion	☐ Other		

Section 10

External Exposures

Location Details

Latitude	[VALUE]
Longitude	[VALUE]
Point where geo coding was taken	Google Earth
Ground Elevation (above mean sea level)	2m
Ground Elevation Source	Munich Re's Risk Suite (NATHAN - Natural Hazard)
Building Finished Floor Elevation (above mean sea level)	At least 0.3m higher than public road based on site observation.

Boundary Exposures

Direction	Exposure Rating	Distance	Comments
North	Low	25m	[LOCATION] Residences (55 storey condominium)
South	Low	15 m	[PROPERTY] 2 (50 storey office building)
East	Low	25 m	Open field
West	Low	120 m	[RELATED PROPERTY] Residences (34 storey condominium)

Natural Catastrophe Exposures

Based on Munich Re's Risk Suite (NATHAN - Natural Hazard)

NATHAN Hazards	Score	Low Hazard	High Hazard
 Earthquake	Zone 0 MM V and below	0	
 Volcanoes	No Hazard	-1	
 Tsunami	No Hazard	-1	
 Tropical Cyclone	No Hazard	-1	
 Extratropical Storm	No Hazard	-1	
 Hail	Zone 2		2
 Tornado	Zone 1 0.1 - 0.5 (per 10,000 km² and year)	1	
 Lightning	Zone 5 20.1 - 40.0 (per km² and year)		5
 River Flood (Defended)	Zone 0 minimal flood risk	0	
 Flash Flood	Zone 4		4
 Storm Surge (Defended)	Zone 100 year return period		100
 Wildfire	No Hazard	-1	

Peril	Source	Exposure Rating	Comments
Pluvial Flood (Heavy Rain / Flash Flood)	MunichRe NATHAN	Low	Localized flash flooding due to heavy rainfall and storm water accumulation is a possibility, with the site situated within a NATHAN flash flood hazard Zone 4 (zones range from 1 to 6, with 6 being the most severe). The site is not listed in the local authorities' list of flood prone areas. Based on site observations, the buildings were at least 0.3m higher than the public road.

Peril	Source	Exposure Rating	Comments
			Based on the above, the risk is deemed as low.
Hail	MunichRe NATHAN	Low	Per Munich Re's NATHAN risk suite, a low frequency and intensity of hailstorms is anticipated at the site.
Tornado	MunichRe NATHAN	Low	Buildings are of robust (reinforced concrete) construction which is expected to provide for wind loading.
Tropical Cyclone	MunichRe NATHAN	No hazard	
River (Inland) Flood	MunichRe NATHAN	Low	Nearest waterway is 65m ([LOCATION]). River flood exposure to this facility is not expected.
Storm Surge / Coastal Flooding	MunichRe NATHAN	No hazard	The site is located about 1.2km from the seashore.
Tsunami	MunichRe NATHAN	No hazard	
Lightning	MunichRe NATHAN	Moderate	Moderate exposure. Per Munich Re's NATHAN risk suite, the site experiences 20-40 lightning strokes per km ² and year. Lightning presents a source of ignition and can cause electrical power surges which may damage and disrupt power supply systems. The building is equipped with lightning protection rods connected to earthing systems with annual testing conducted.
Earthquake	MunichRe NATHAN	Low	Low exposure. Singapore is not situated within an earthquake zone. However, Singapore do experience occasional tremors due to earthquake that happens in the region. Per Munich Re's NATHAN risk suite, the site is situated within an area with a probable maximum earthquake intensity of Modified Mercalli Intensity (MMI) V and below based on a return period of 475 years for medium subsoil conditions.

Peril	Source	Exposure Rating	Comments
Volcanic	MunichRe NATHAN	No hazard	
Wild Fire	MunichRe NATHAN	No hazard	
Other	MunichRe NATHAN	Low	For underground soil condition, it is classified as soft rock/dense soil and there is no signs of soil subsidence at time of visit.

Section 11

Security

Site Perimeter

Vehicle Access	Vehicles hourly and season parking are allowed in this site. Typical entry gantry with CCTV monitoring provided.
Perimeter Fencing	Not provided.
Entry Provisions Posted	Security guard station at the building main entrance and other strategic locations.
No Trespassing Signage	Not provided.
Fencing Height	Not provided
Exterior Lighting	Along the perimeter

Building Physical Features

Ground Level Windows Secure	Full height glass panel wall on ground level. Entrance to the buildings are by access control.
Lock and Key Control	Each area is provided with access control via staff / visitor / contactor, security badge with differentiation in access based on needs basis. Office and retail tenants will provide their own access control in their respective units, which is lock & key or access card system.

Security Staff / Guards

On-Site Guards	24/7 security for the development is provided by contract guards from a third party security agency, Prosegur. Guards are unarmed and perform 12 hour shifts, with 7 guards during the day shift and 6 guards in the night shift. Guards are deployed at the security counters, access points and common areas of the development.
Visiting Patrols	All security personnel have gone through proper training stipulated by the local regulatory authority.
Police Surveillance	Regular patrol is expected to be conducted by the local police in the vicinity.
Recorded Rounds	Regular security patrols are carried out for the entire development including the building perimeters, common

Security Staff / Guards

	areas, roofs and M&E plant rooms. An electronic clocking system is utilised during patrols, with a total of 193 installed clocking points.
Rounds Frequency	Day: 4 physical clocking Night: 4 physical clocking
CCTV	Around 500 CCTV cameras have been installed throughout the development. These cover both internal and external areas including the main building lobbies, car parks, building perimeters, lift lobbies and interiors of lift carriages. CCTV footage is recorded via a DVR system at the FCC, with storage capacity allowing for at least 30 days of footage to be stored/played back. CCTV cameras covering critical areas (e.g. entrance points) are equipped with video analytics capabilities.

Intruder Detection

Motion Sensors	CCTV cameras are equipped with Video Content Analysis (VCA), allowing constant monitoring of certain areas for intrusion, slip & fall, illegal parking, etc.
Beams	Not provided
Door Contacts	Yes
Access Monitored	Yes, by CCTV.
ID Cards / Badges	Yes
Visitors Monitored	Yes, all contractors are required to change passes before commencement of work. Visitors are only given access after registering at the reception area.

Other Security Programs / Controls

Emergency Response	Documented emergency response plan has been established. Company Emergency Response Team (CERT) is available during operational hours. Each CERT team has 5 first responders. All CERT members had undergone relevant training according to their appointed roles. Some of the training include industrial firefighting/rescue and first aid training by SCDF-certified third party training provider. Drills and table top exercises are conducted 2 x per year
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Other Security Programs / Controls

Employee Screening	Expected to be conducted as part of the employment process.
Safe / Vaults	Not expected. Tenants may have their own safe / vaults for cash management.
Mobile Equipment Storage	Not expected.
Bomb Threat	Any explosive materials are strictly controlled in Singapore. Exposure is considered remote.
Cash / Check Management	Minimal cash kept on site. Tenants are expected to manage their cash transactions from their operations.
Parking	The car park is open to the public and overnight parking is permitted

Appendix A

Site Drawing

[PROPERTY] – L1 Floor Plan

[PROPERTY] Typical Floor Plan

Appendix B

Sprinkler Water-proving Test Results

SPRINKLER PUMP WATER-PROOFING TEST(IF APPLICABLE)

CONTROL VALVE NO.	LOCATION OF CONTROL VALVE	AREA SERVED	HAZARD GROUP	FLOW RATE(L/min)	MINIMUM RUNNING PRESSURE(KPA)	ACTUAL FLOW RATE(L/min)	ACTUAL RUNNING PRESSURE(KPA)
CV- B3	B3	B3	OH 3	1100 1350	195 165	1100 1350	450 430
CV- B2	B2	B2	OH 3	1100 1350	195 165	1100 1350	450 430
CV- B1	B1	B1	OH 3	1100 1350	200 170	1100 1350	430 410
CV- L1	L1	L1 & Mezzanine	OH 3	1100 1350	215 185	1100 1350	370 360
CV- 02	L2	L2	OH 3	1100 1350	215 185	1100 1350	330 310
CV- 03	L3	L3	OH 3	1100 1350	215 185	1100 1350	500 480
CV- 04	L5	L5	OH 1	375 540	145 115	375 540	620 600
CV- 05	L6	L6	OH 1	375 540	149 119	375 540	480 470
CV- 06	L7	L7	OH 1	375 540	166 136	375 540	440 430
CV- 07	L8	L8	OH 1	375 540	135 105	375 540	360 350
CV- 08	L9	L9 & L10	OH 1	175 540	180 150	175 540	350 340
CV- 09	L11	L11 & L12	OH 1	375 540	180 150	375 540	250 240
CV- 10	L11	L11 & L14	OH 1	375 540	180 150	375 540	940 930
CV- 11	L15	L15 & L16	OH 1	375 540	180 150	375 540	860 850
CV- 12	L17	L17 & L18	OH 1	375 540	180 150	375 540	900 890
CV- 13	L19	L19 & L20	OH 1	375 540	180 150	375 540	670 660
CV- 14	L21	L21 & L22	OH 1	375 540	180 150	375 540	560 550
CV- 15	L23	L23 & L24	OH 1	375 540	180 150	375 540	750 740
CV- 16	L25	L25 & L26	OH 1	375 540	180 150	375 540	6900 680



A handwritten signature in blue ink, appearing to be "Lee Leong Ying".

SPRINKLER PUMP WATER-PROOFING TEST(IF APPLICABLE)

CONTROL VALVE NO.	LOCATION OF CONTROL VALVE	AREA SERVED	HAZARD GROUP	FLOW RATE(L/min)	MINIMUM RUNNING PRESSURE(KPA)	ACTUAL FLOW RATE(L/min)	ACTUAL RUNNING PRESSURE(KPA)
CV- 17	L27	L27 & L28	OH 1	375 540	180 150	375 540	790 780
CV- 18	L29	L29 & L30	OH 1	375 540	180 150	375 540	680 670
CV- 19	L31	L31 & L32	OH 1	375 540	180 150	375 540	620 550
CV- 20	L33	L33 & L34	OH 1	375 540	180 150	375 540	510 450
CV- 21	L35	L35 & L36	OH 1	375 540	180 150	375 540	410 400
CV- 22	L37	L37 & L38	OH 1	375 540	180 150	375 540	340 310
CV- 23	L39	L39 & L40	OH 1	375 540	180 150	375 540	780 770
CV- 24	L41	L41 & L42	OH 1	375 540	180 150	375 540	720 710
CV- 25	L43	L43 & L44	OH 1	375 540	180 150	375 540	670 660
CV- 26	L45	L45 & L46	OH 1	375 540	184 154	375 540	560 550
CV- 27	L47	L48 & L48	OH 1	375 540	218 188	375 540	490 480



Appendix C

Photo

FCC at L1	Engineering store
Sprinkler Pump	EV charging bay
Back-up air-cooled chillers	Emergency Power Gen-set
Engineering Store	Transformer room
Typical Electrical Room	Interconnection to Tower 2 with fire shutter installed



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